

Computer Engineering Department
Program: Sem VI
Course: Cloud Computing Lab(CSL605)
PART A

(PART A: TO BE COMPLETED BY STUDENTS)

Experiment No.5

A.1 Aim:

To demonstrate and implement Storage as a service using AWS S3 Service

A.2 Prerequisite:

Knowledge of Networking, Distributed Computing and knowledge of Software architectures.

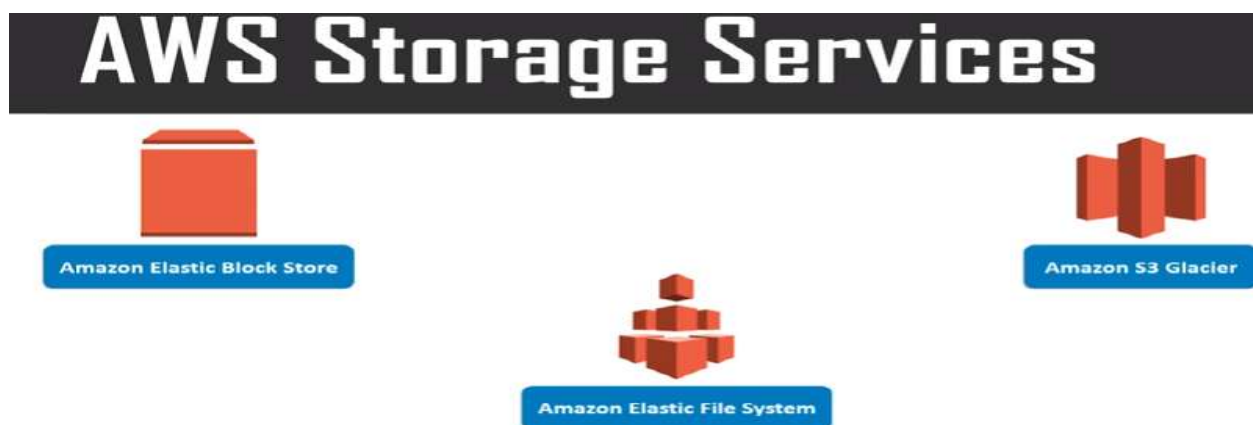
A.3 Objective:

Objectives this experiment is to provide students overview of AWS storages, its Features and Services.

A.4 Outcome: (LO3)

After successful completion of this experiment student will be able to; Implement IAAS deployment model of cloud.

A.5 Theory:



Amazon Elastic Block storage (EBS): Storage in the form of blocks, each of these blocks associated with one particular instance, so when to access this block make sure that you have an instant connected to it and storage is accessed through that instant only.

Amazon Elastic File System (EFS): It is shared file system; hence it is not attached to a particular system or operating system.

Amazon S3 Glacier: Need to store archive data, certain data that we would not want to retrieve or access on daily basis or frequently, such data you can put on **S3 Glacier**. That is it locks data for certain time during which you cannot access it, once you clear that duration you are free to access that data.

This storage is very cheap as compared to other data storage.

Example: Hospital System (birth certificate data): Birth certificate given by hospital once

baby takes birth, so once you get it you are not requesting it again and again and hospital need to maintain it. That is data which is important but not needed on daily basis.

AWS Storage Gateway: Act as middle ware to move data from one system to another system.

What is Amazon S3 (Simple Storage Service)

- 1) Amazon s3 has a simple web service interface that you can use to store and retrieve any amount of data, at anytime from anywhere on the web.
- 2) **S3 works on objects and buckets** (Bucket is an container and an object (doc, image, file etc) is an file, which you can stored in container)
- 3) Online cloud storage

Pre-S3

Online Cloud Storage

- Online Storage – Upload files, folders, images, songs, videos from a machine and access it from anywhere in the world.



Dropbox



iCloud



OneDrive

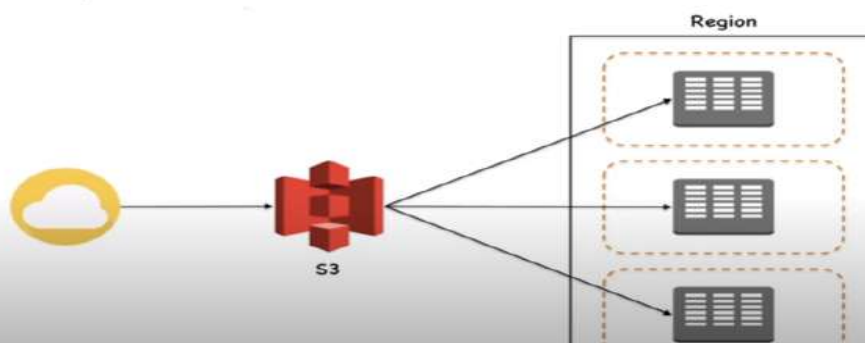


Google Drive

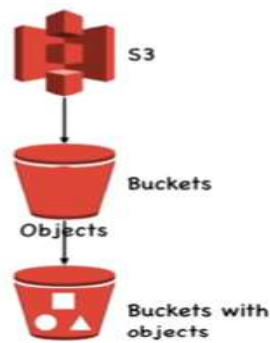
S3 Concepts

Simple Storage Service

- Simple Storage Service is storage for internet.
- S3 provides web service which can be used to store and retrieve unlimited amount of data. Same can be done programmatically using Amazon provided APIs.
- **S3 Data Consistency Model**
 - S3 provides highly durable and available solution by replicating all data in multiple data centers in a region.
 - Data uploaded in a particular region never leaves it.
 - Read-after-write consistency.
 - Eventual consistency.



- S3 follows a storage hierarchy in keeping data (documents, images, videos, files etc.).



- Management console or S3 APIs can be used to manage buckets and objects.
- Bucket names have to be Globally unique irrespective of which region they are created in.
- Max 100 buckets can be created per account.

Amazon S3 (Simple Storage Service) provides object storage which is built for storing and recovering any amount of information or data from anywhere over the internet



4)

- ✓ Amazon S3 provides storage through web services interface
- ✓ It is designed for developers where web-scale computing can be easier for them
- ✓ It provides 99.999999999% durability and 99.99% availability of objects
- ✓ It can store computer files up to 5 terabytes in size

5)

Object & Bucket in Amazon S3

An object consists of data, key(assigned name) and metadata

A bucket stores objects

When data is added to the bucket, Amazon S3 creates a unique version ID and allocates it to the object

For Example:



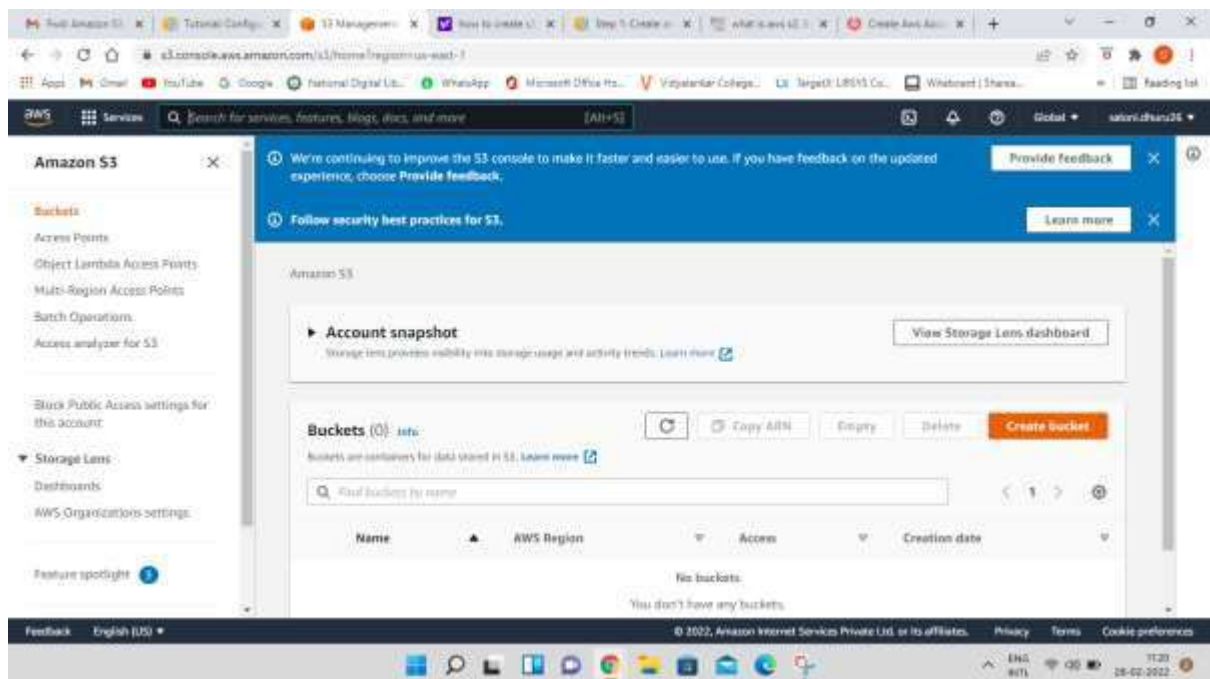
Object: folder/Penguins.jpg
 Bucket: simplilearn
 Link Address: <https://s3.amazonaws.com/simplilearn/folder/Penguins.jpg>

Key(name)

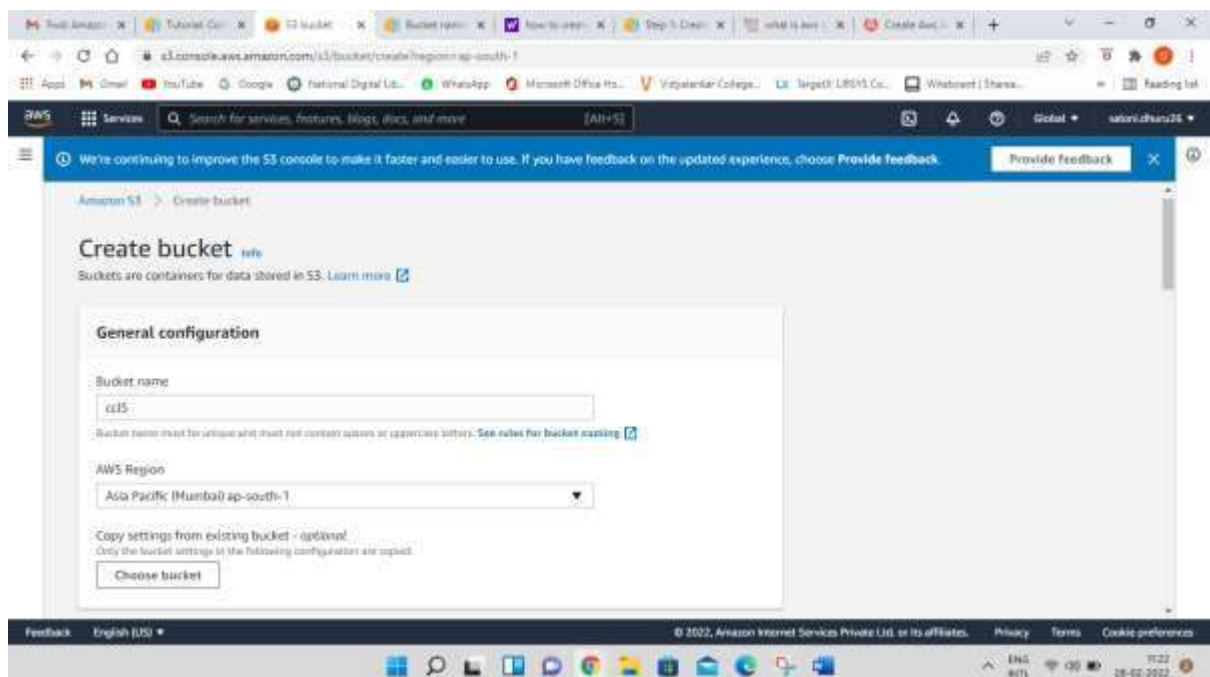
Version ID

Steps to Implement Storage as a Service using Own Cloud/ AWS, Glaciers

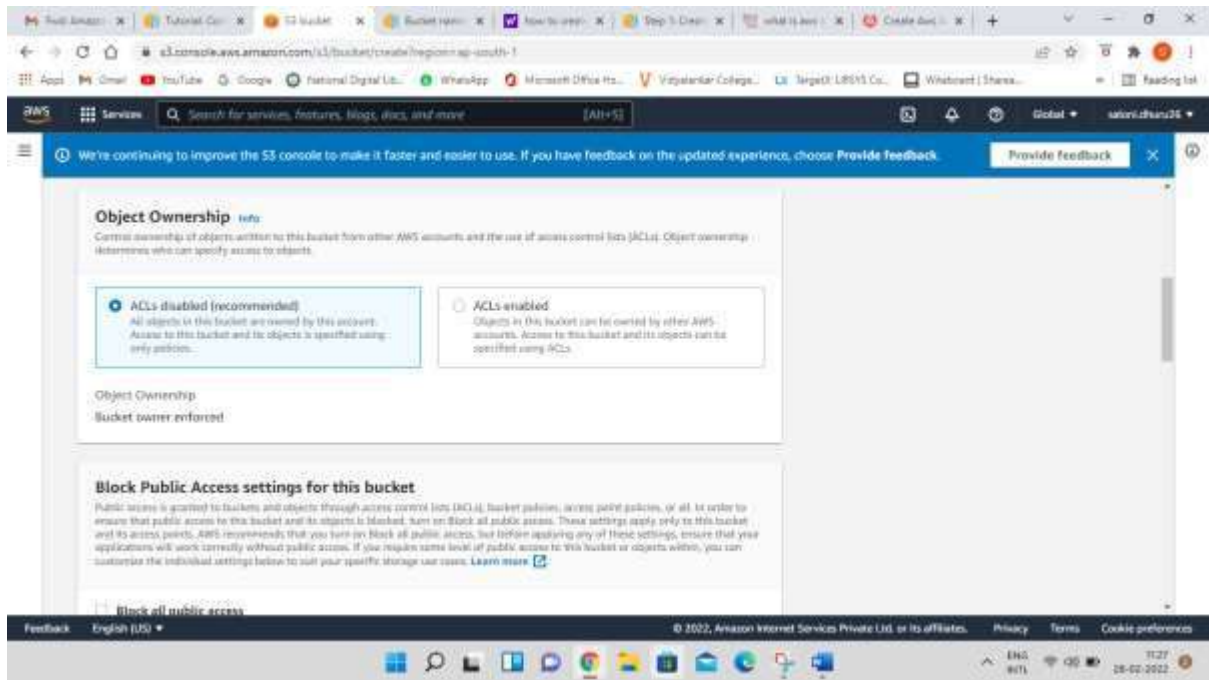
Step-1: click on create bucket



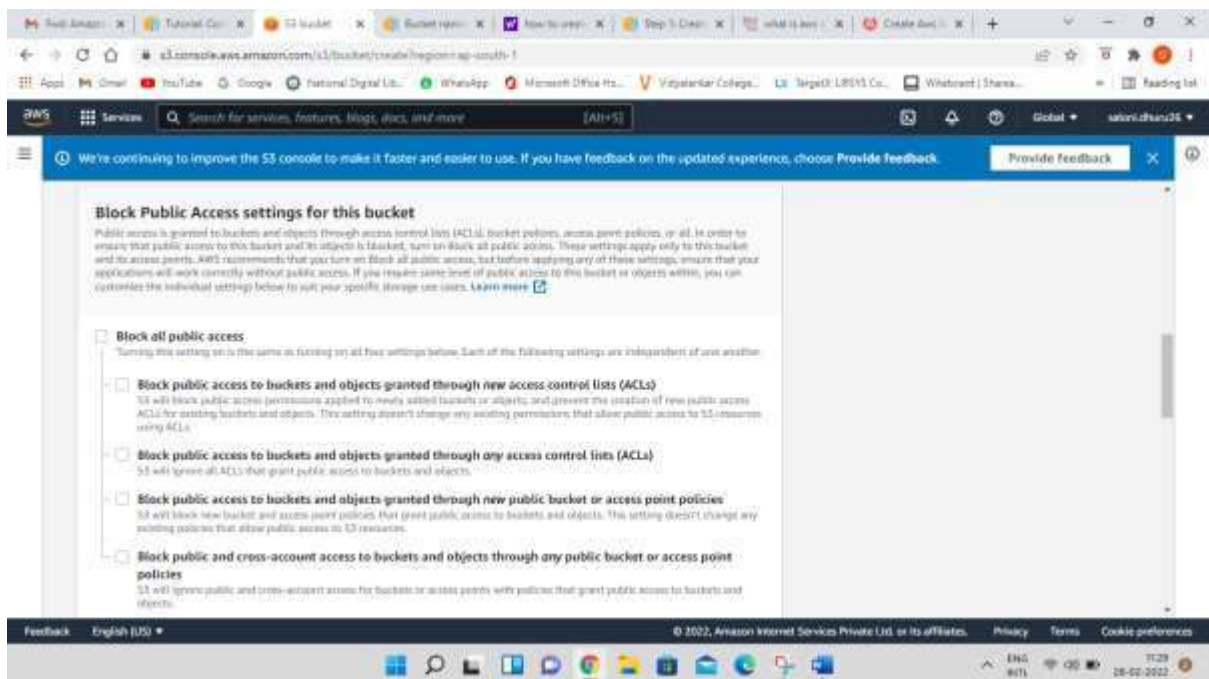
Step-2: Give Bucket name & select region for storage



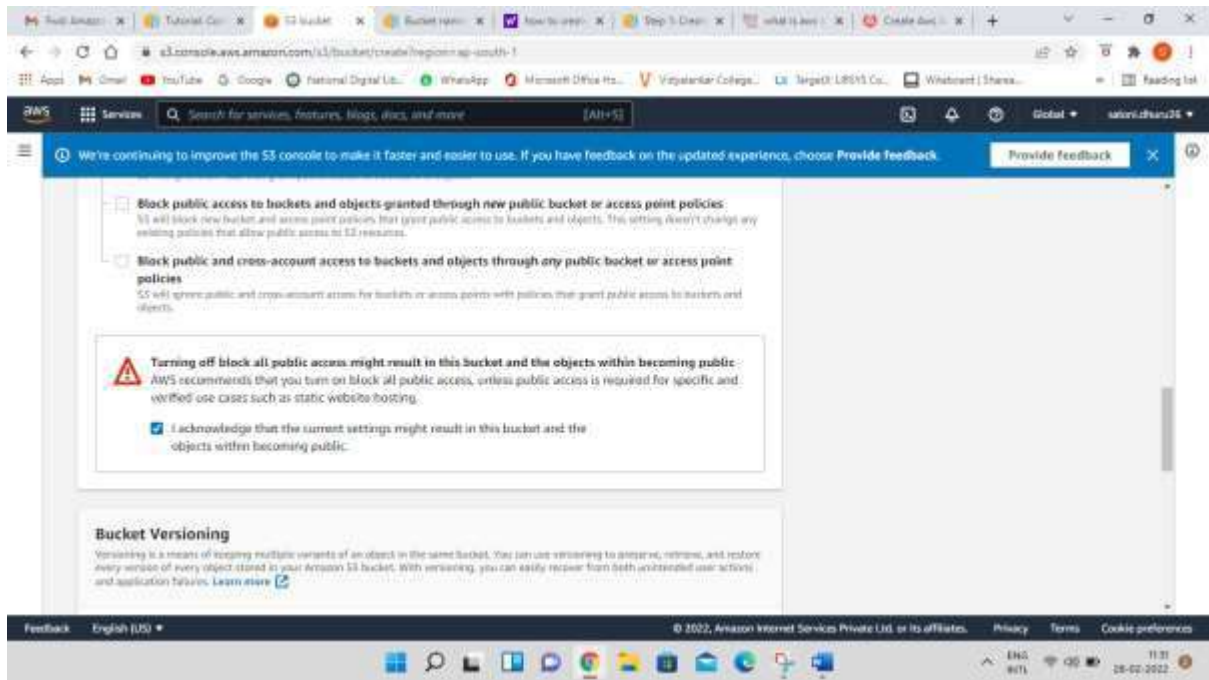
Step-3: Keep object ownership setting as ACLs Disabled as by-default



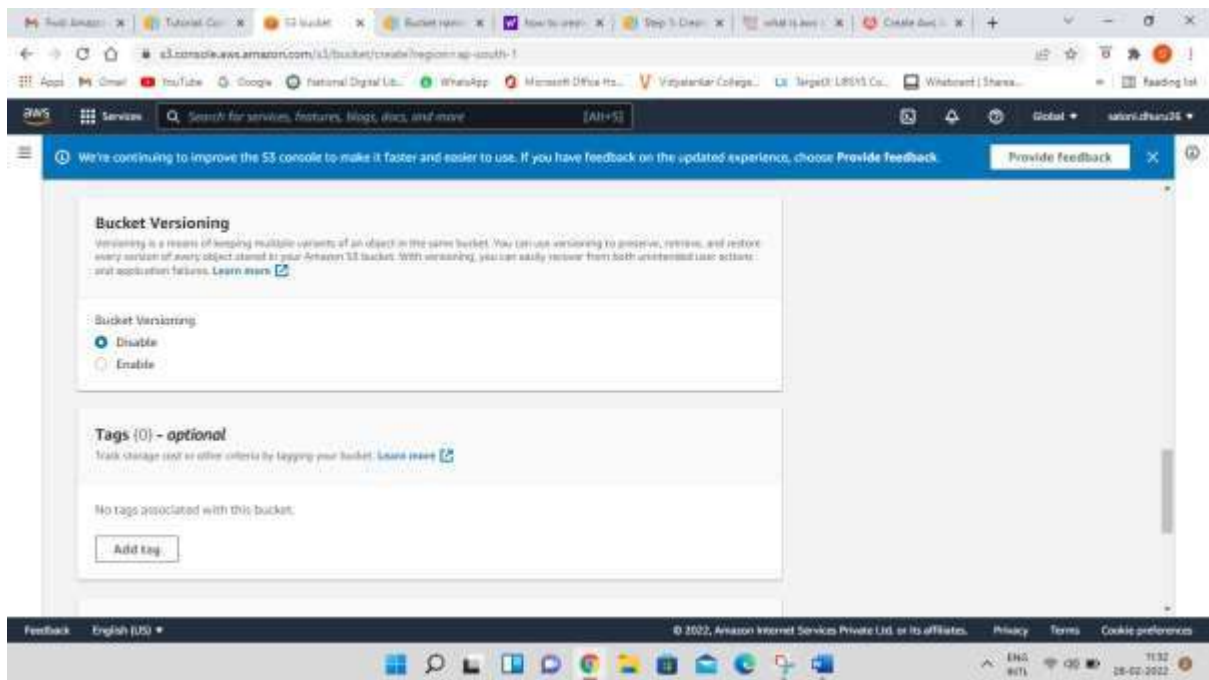
Step-4: Disable block all public access checkbox



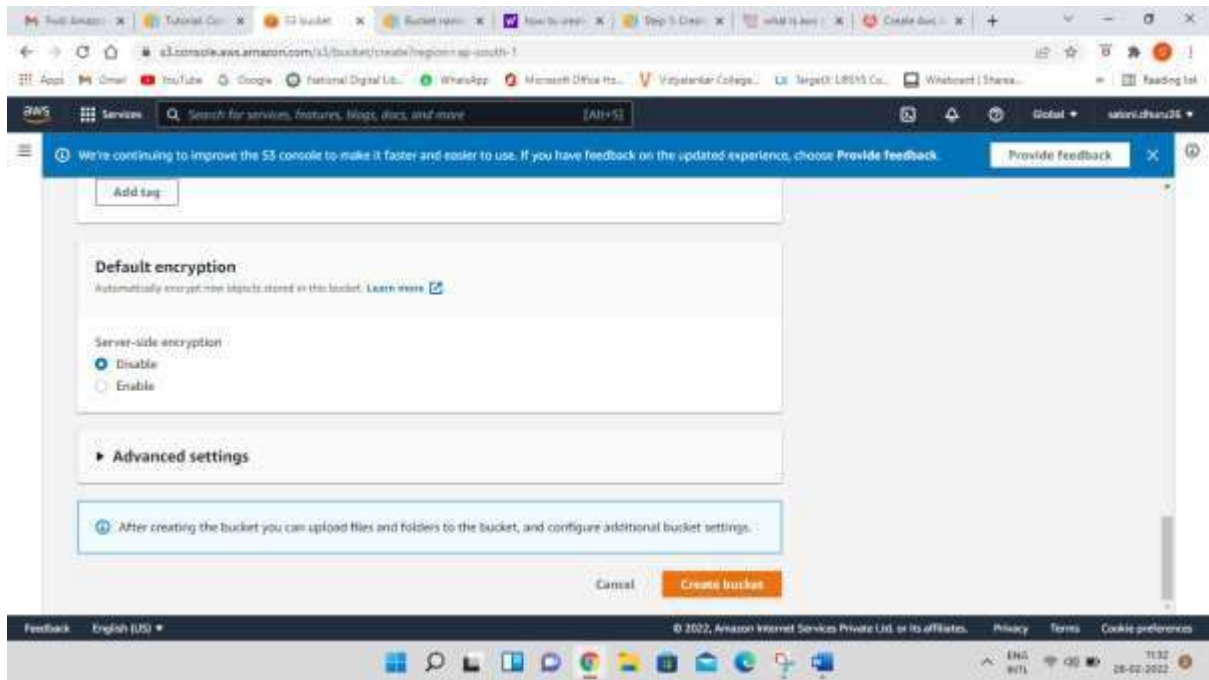
Step-5: Select the checkbox for Turning off block all public access might result in this bucket and the objects within becoming public



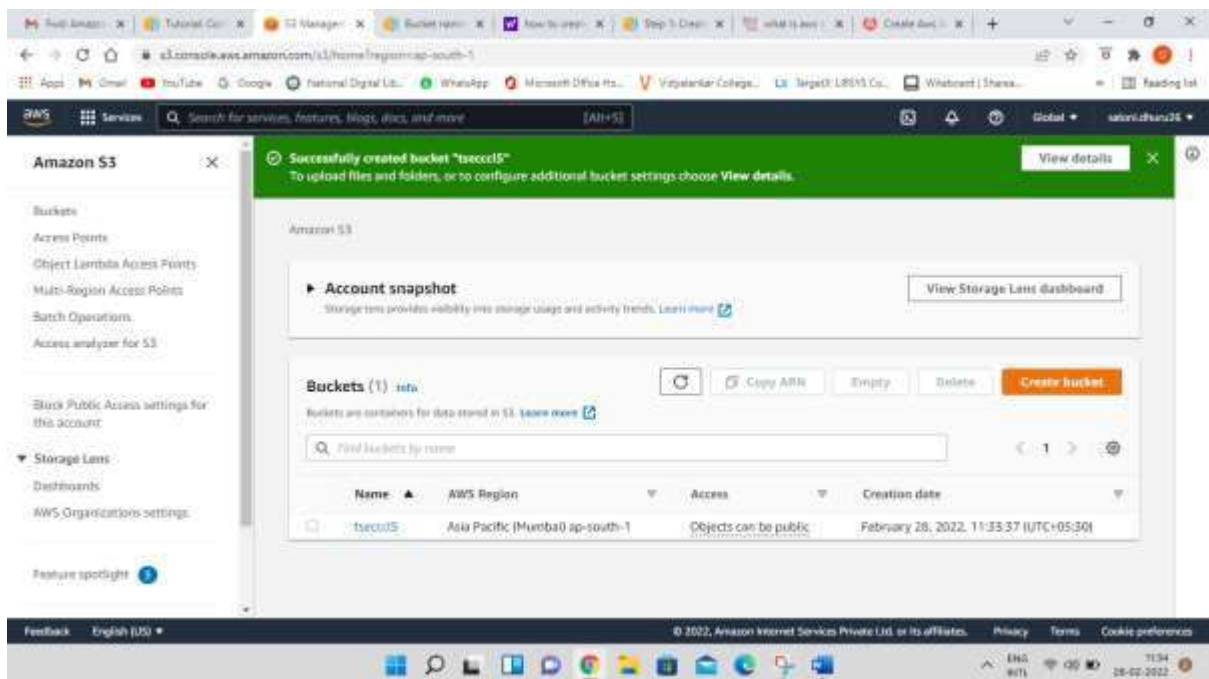
Step-6: Keep bucket versioning as disabled and add tags if required.



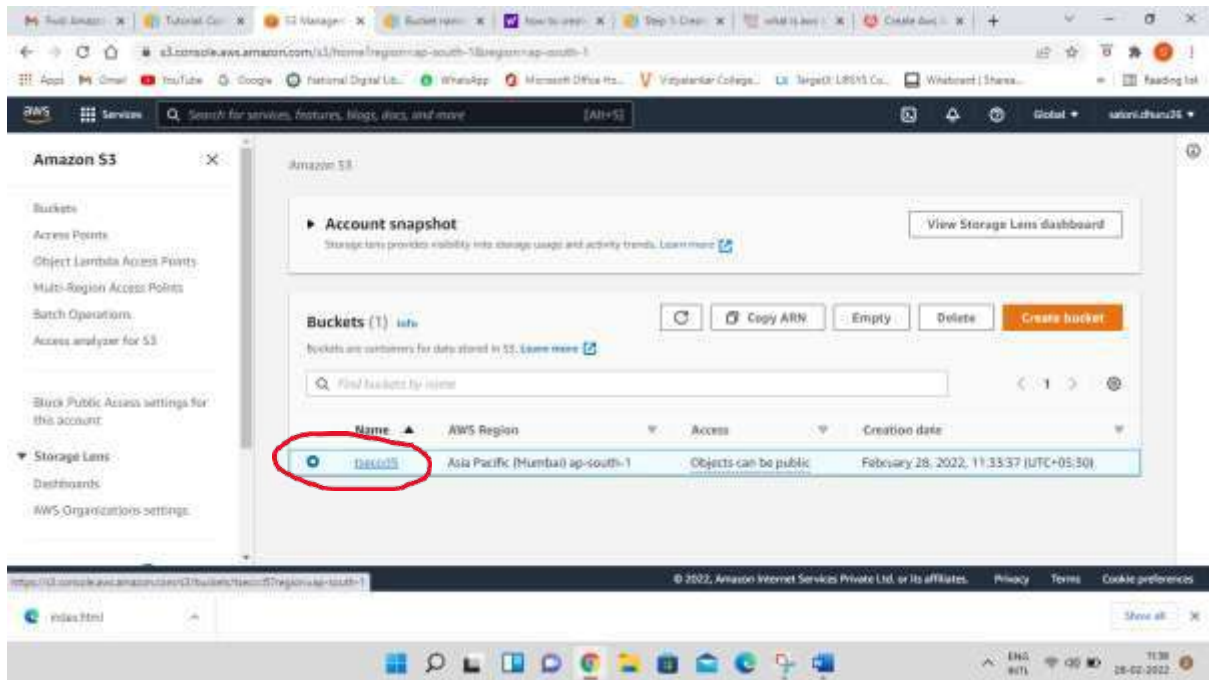
Step-7: Keep default encryption disabled and click on create bucket button



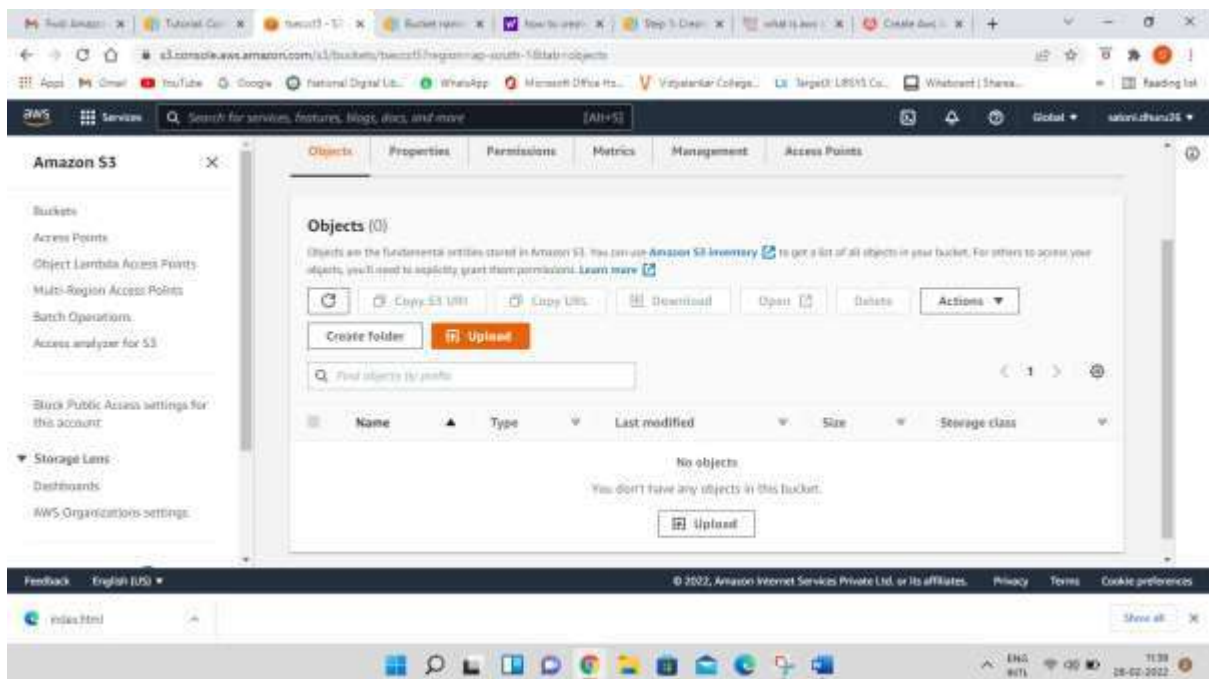
You can now see the successful creation of your bucket



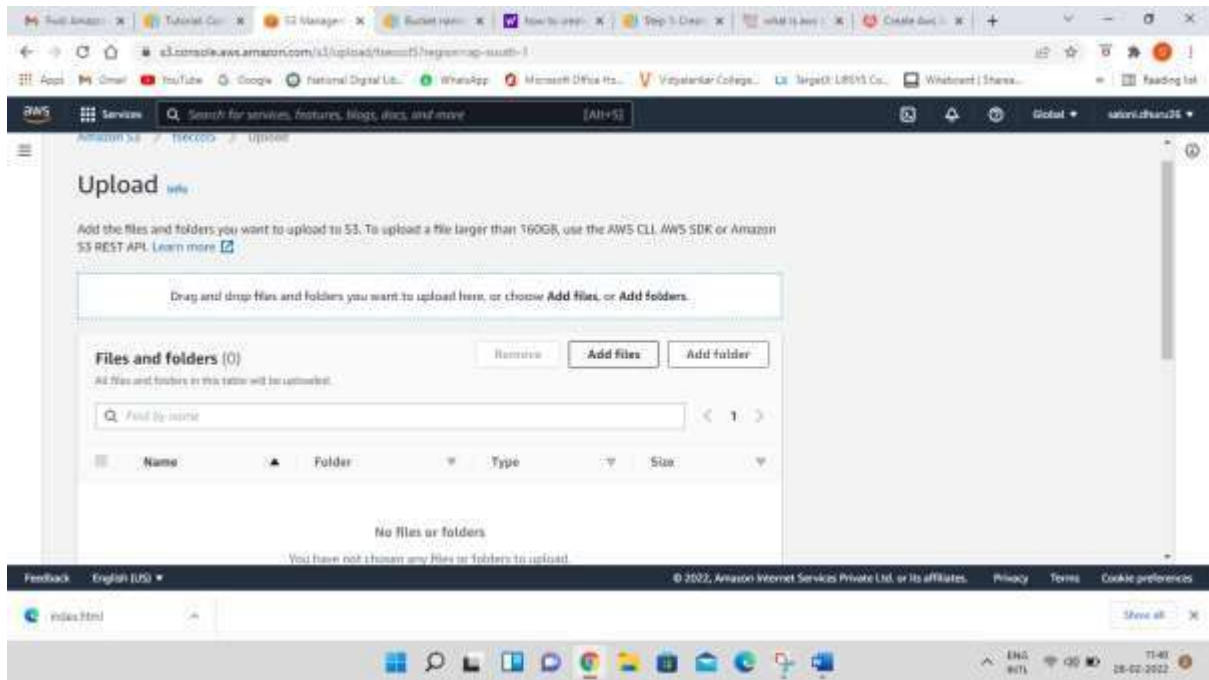
Step-8: now click on the bucket that you have created



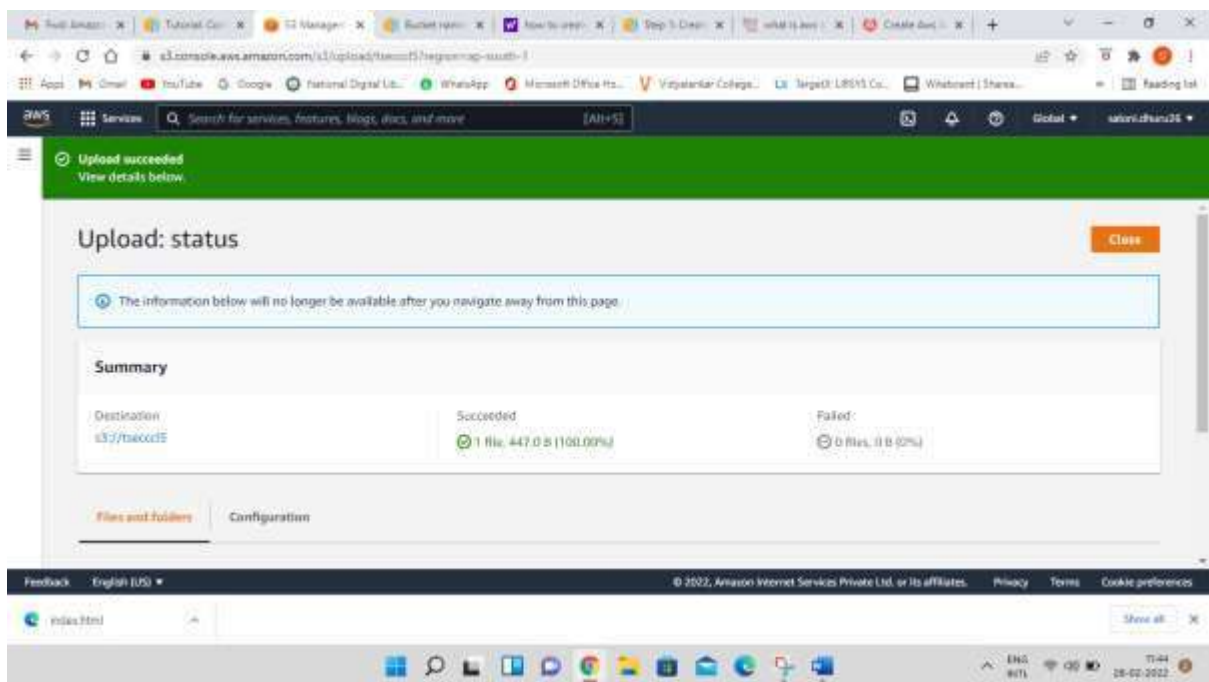
Step-9: You can either create a folder here or upload an existing file in the bucket



Step-10: now click on upload button and click on add files button browse your local machine and select which file you need to upload on S3 next click on upload button at bottom right end

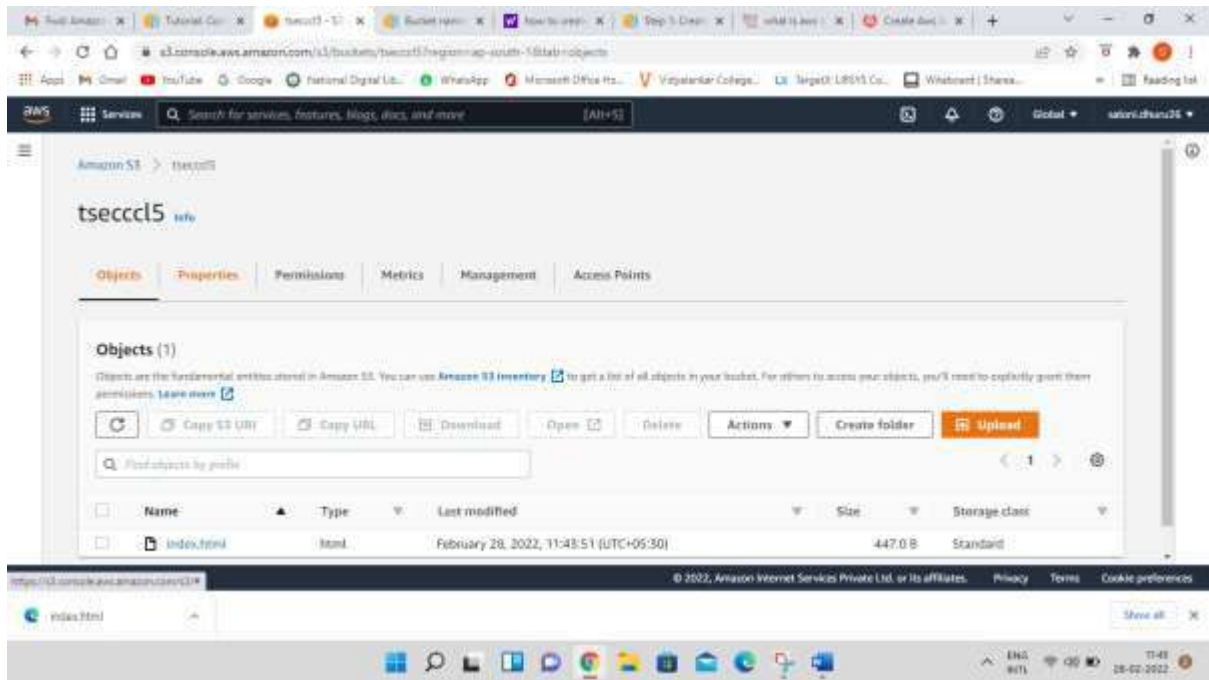


Now you can check the upload status screen

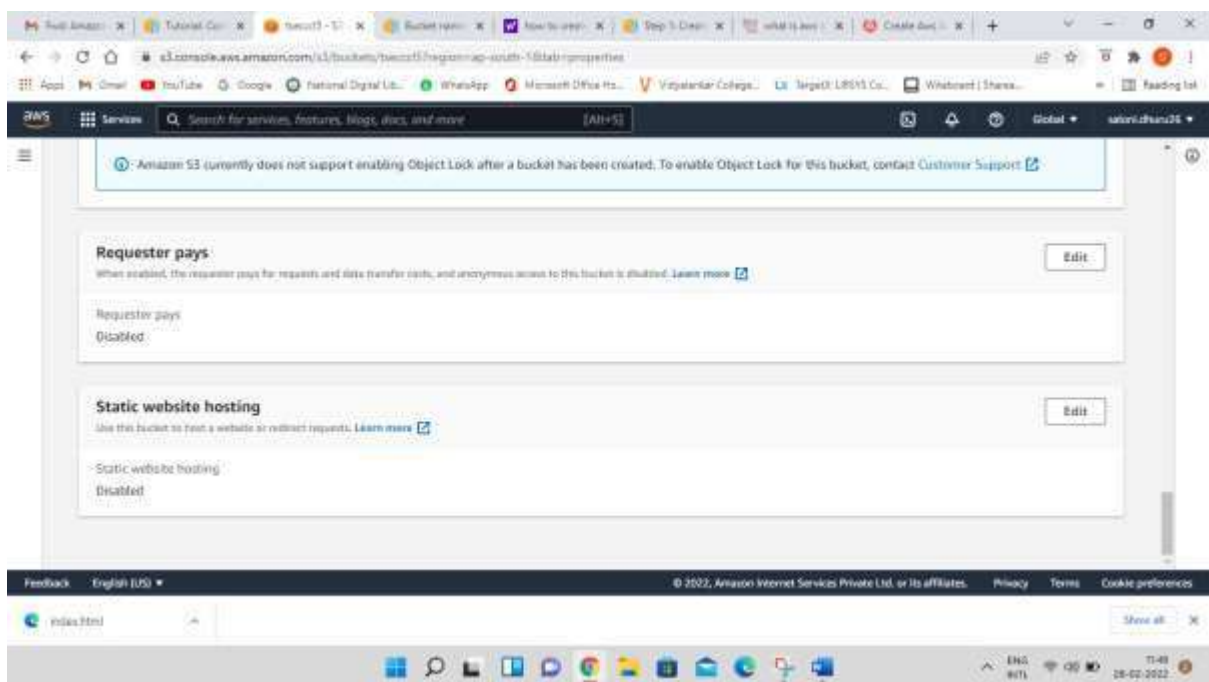


Now click on close button

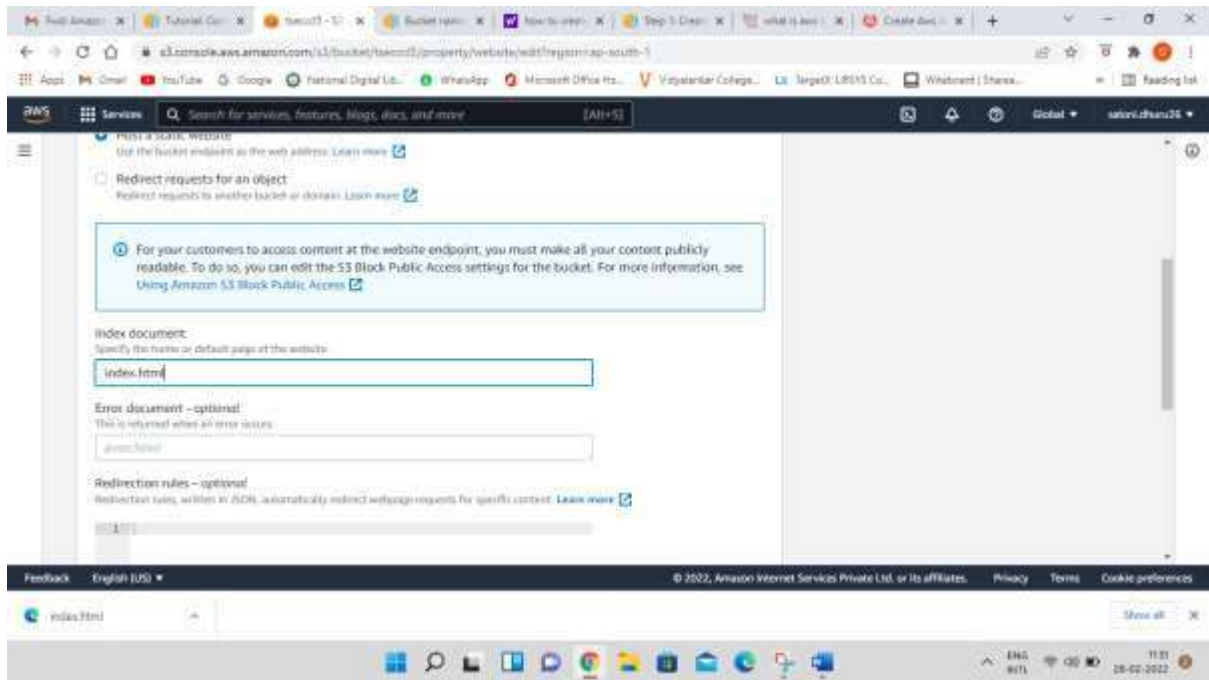
The screen will appear as below



Step-11: Select properties and scroll down to **Static website hosting** option which is disabled now click on Edit option on right side



Step-12: Enable the radio button and specify the file name in **Index document** which you have added in S3



Edit static website hosting [Info](#)

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

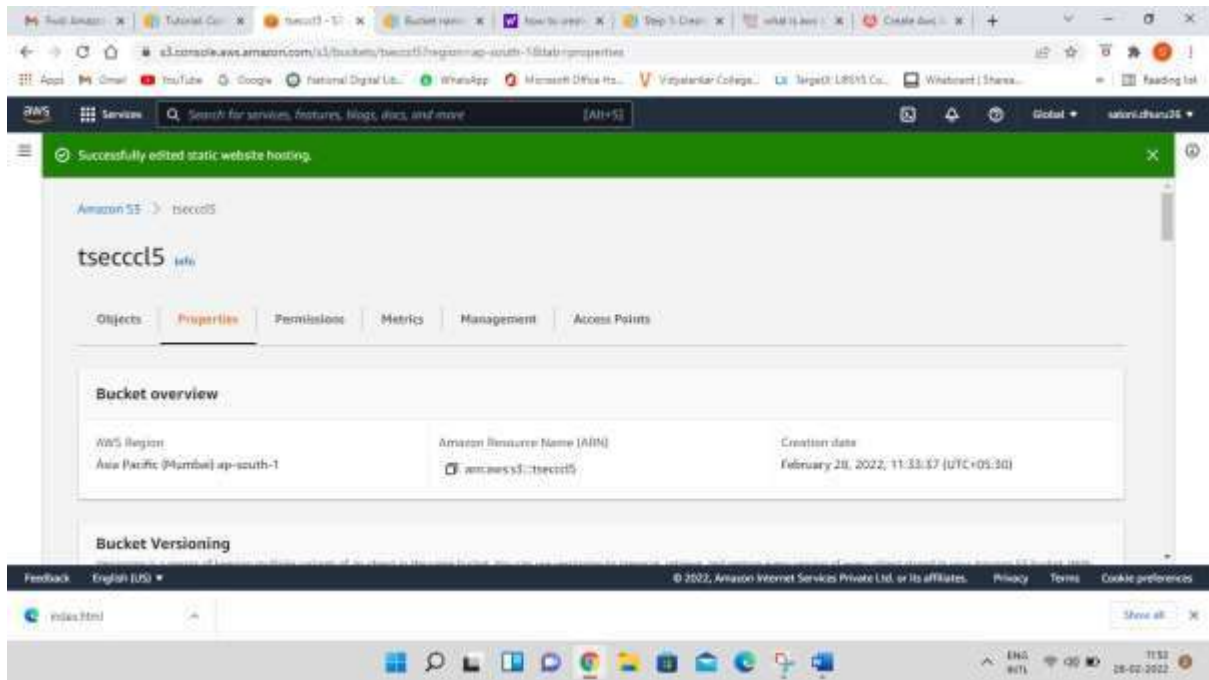
- ☐ Disable
- ☒ Enable

Hosting type

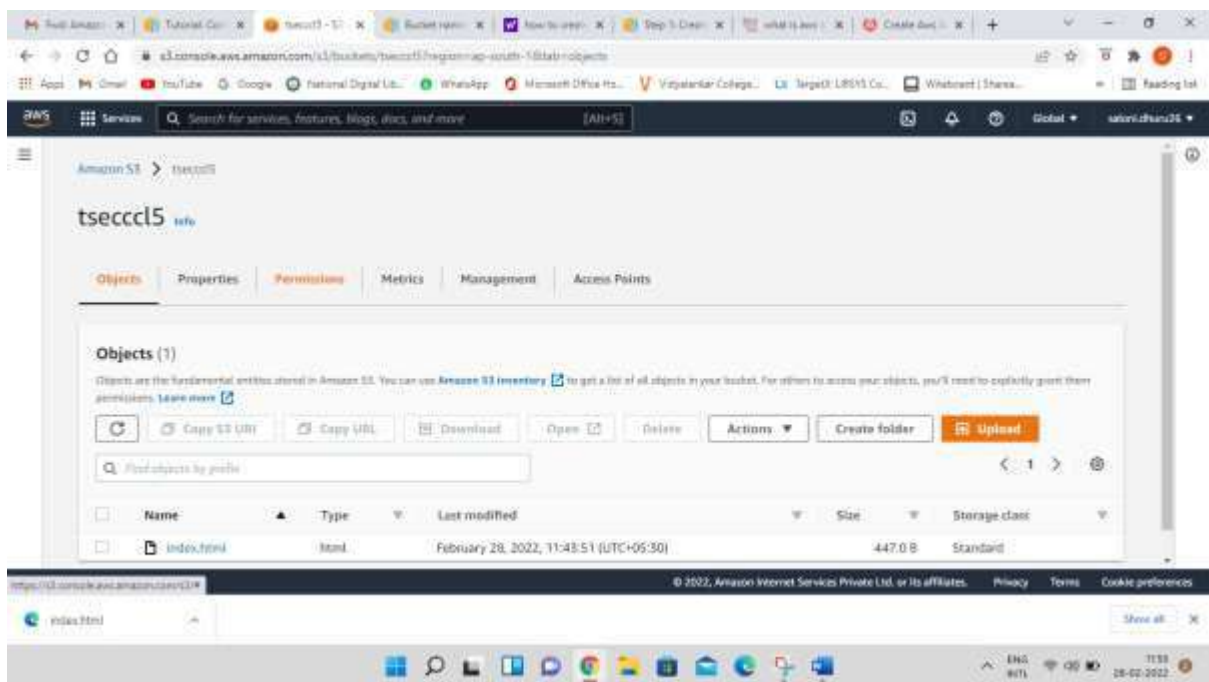
- ☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)
- ☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

Scroll down and save the changes at bottom right

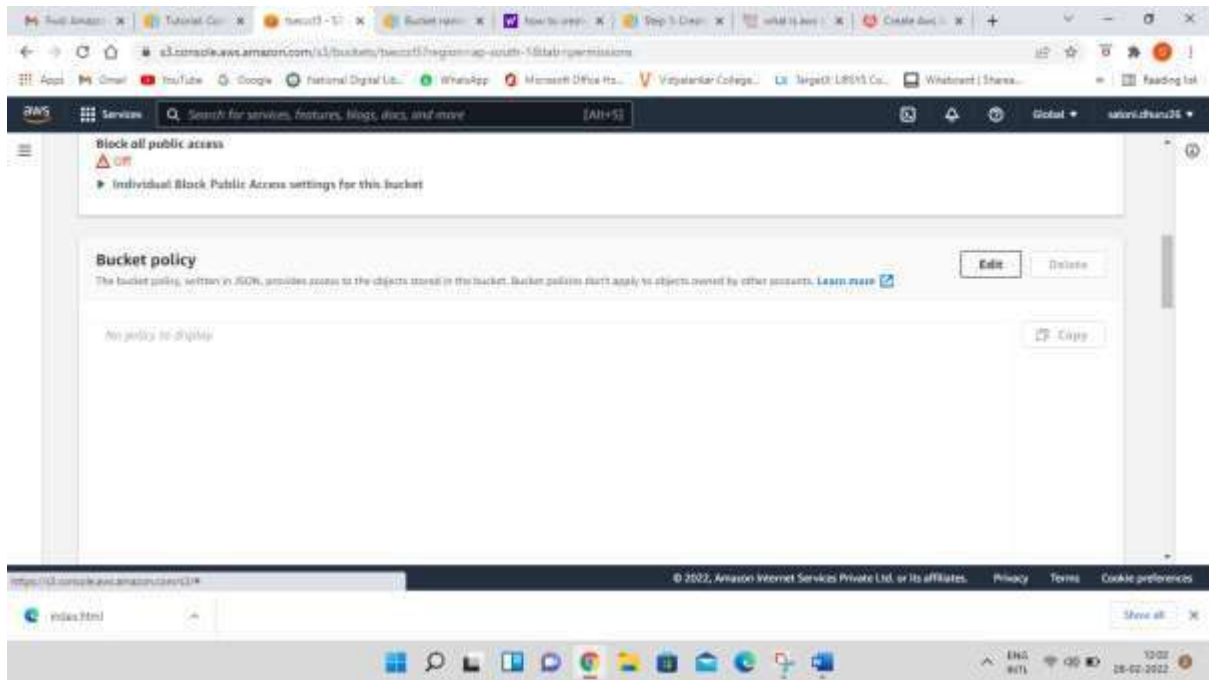
Following screen will appear



Step-13: Click on Permissions Tab



Step-14: In **bucket policy** click on Edit option

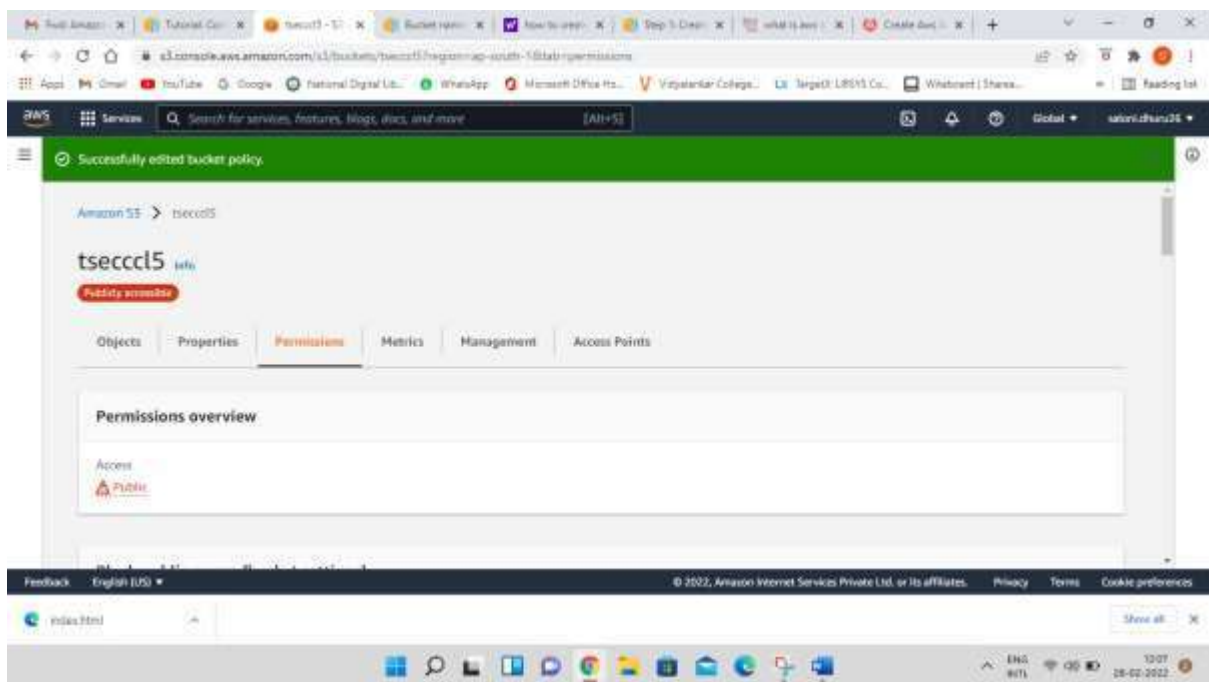
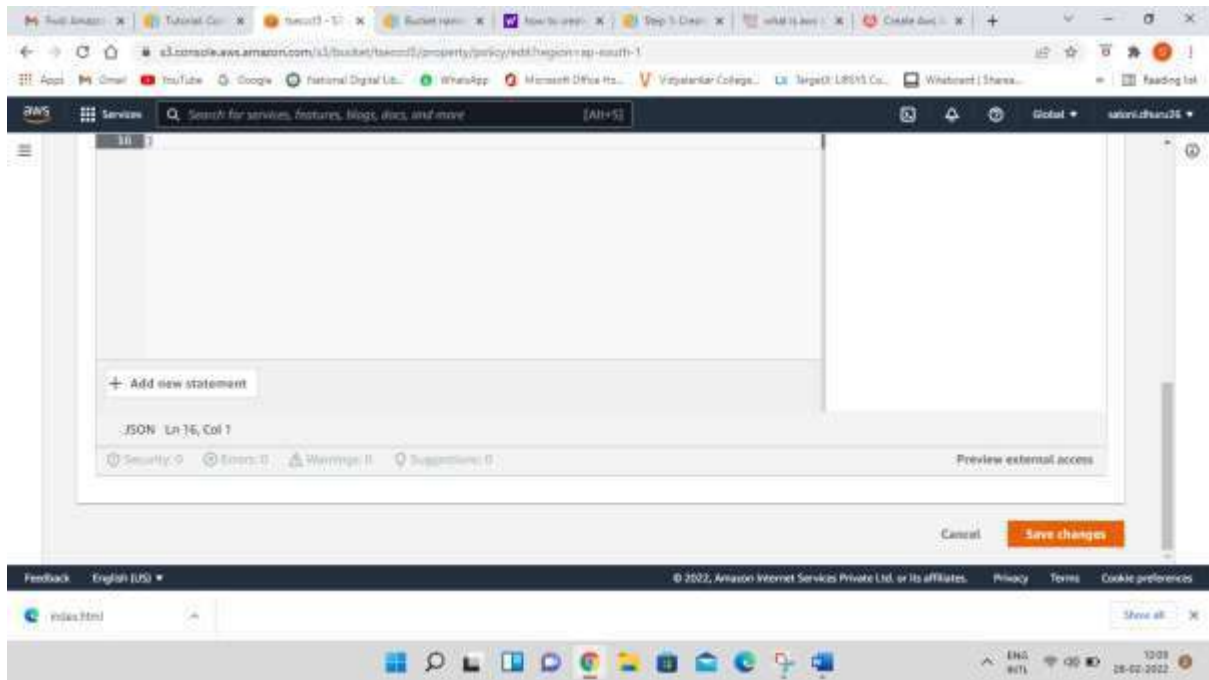


Step 15- after clicking on edit button paste the following code in bucket policy

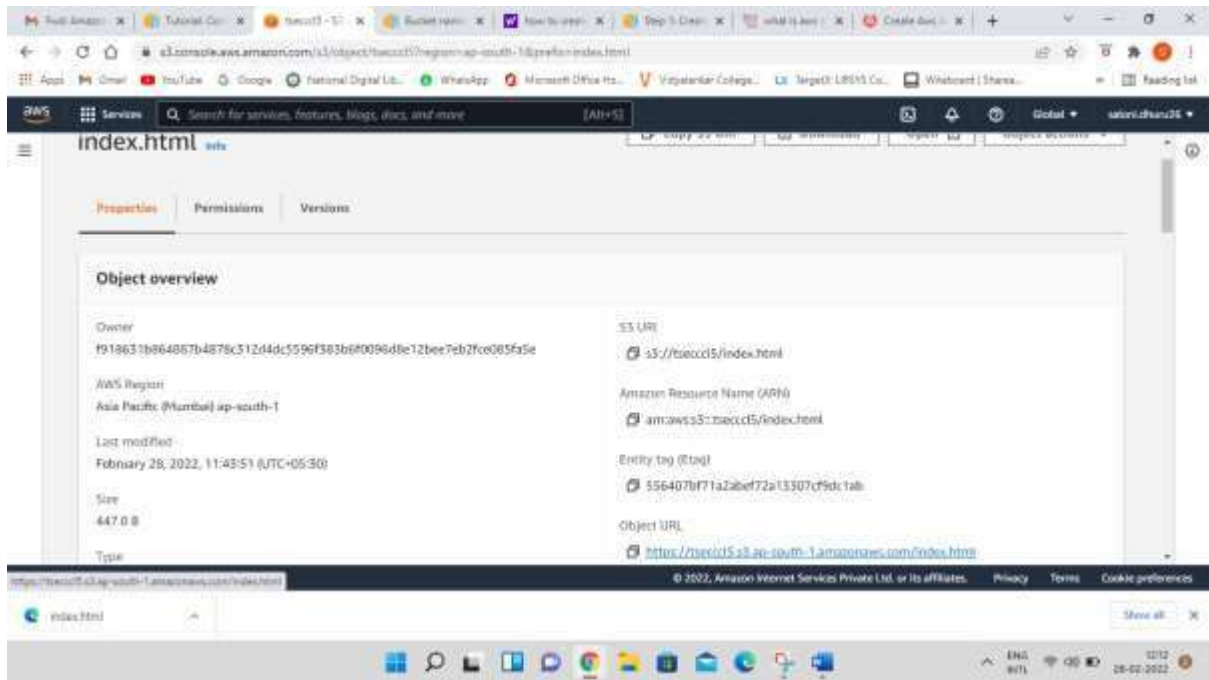
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::Bucket-Name/*"
      ]
    }
  ]
}
```

Note-Make sure that you add your bucket name in the code above

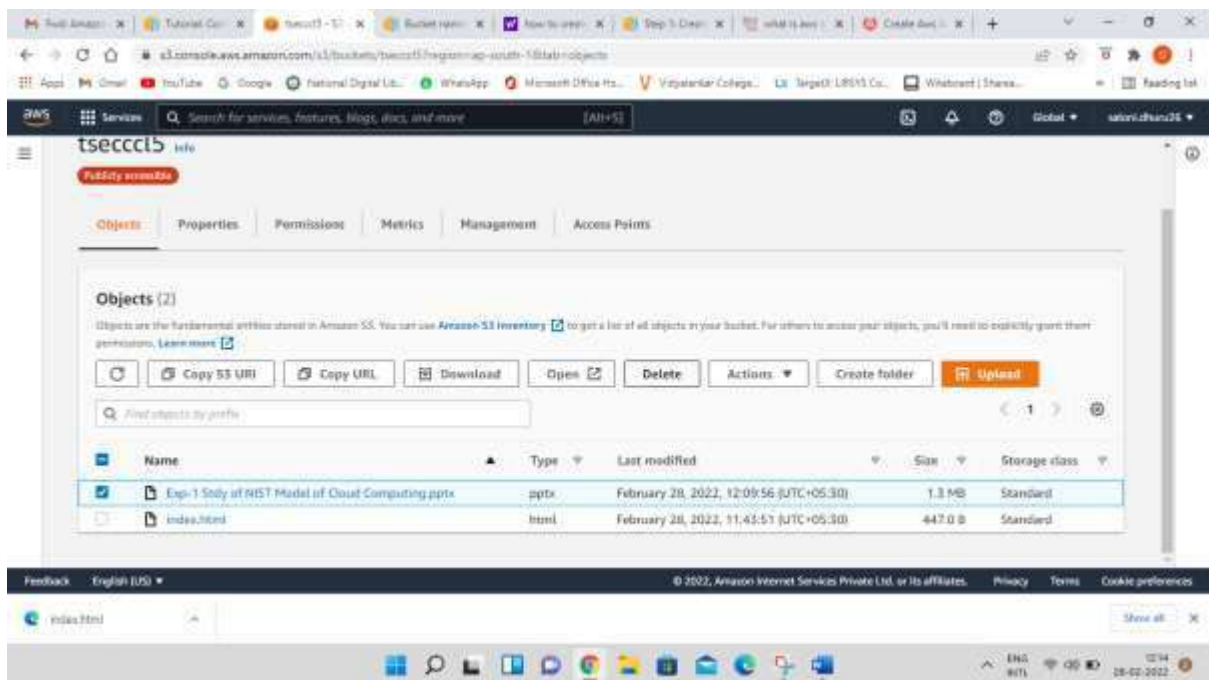
Scroll down and click on Save Changes button



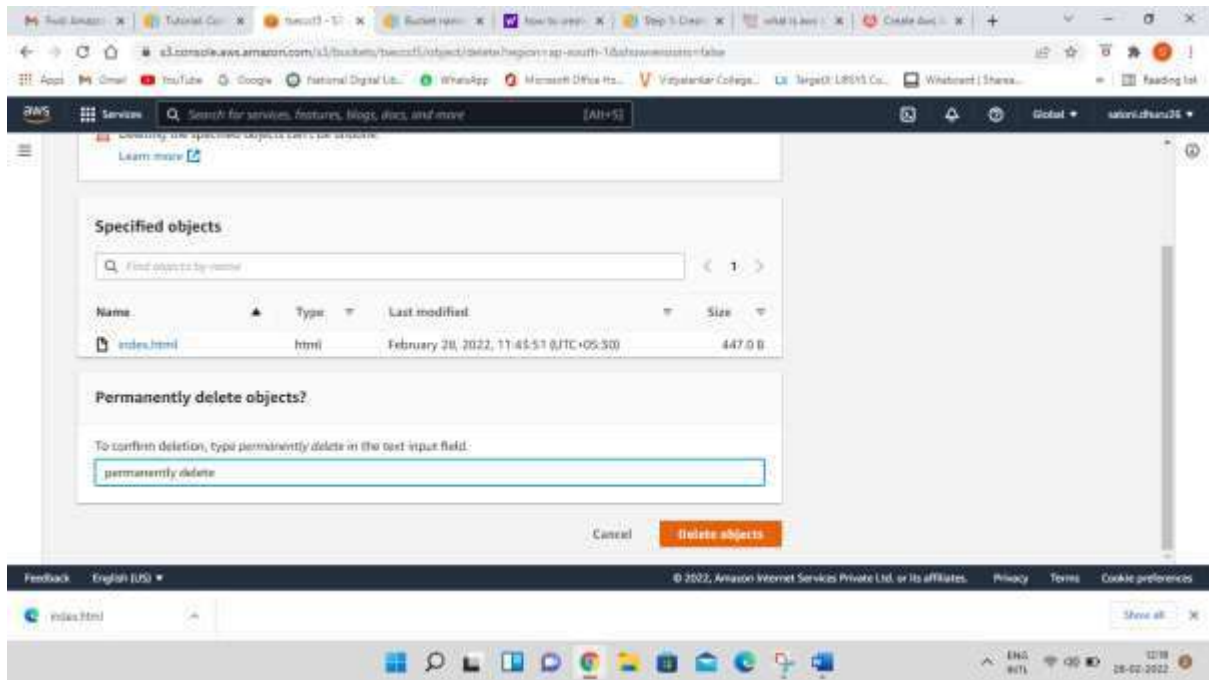
Step-16: open your html file and click on Object URL



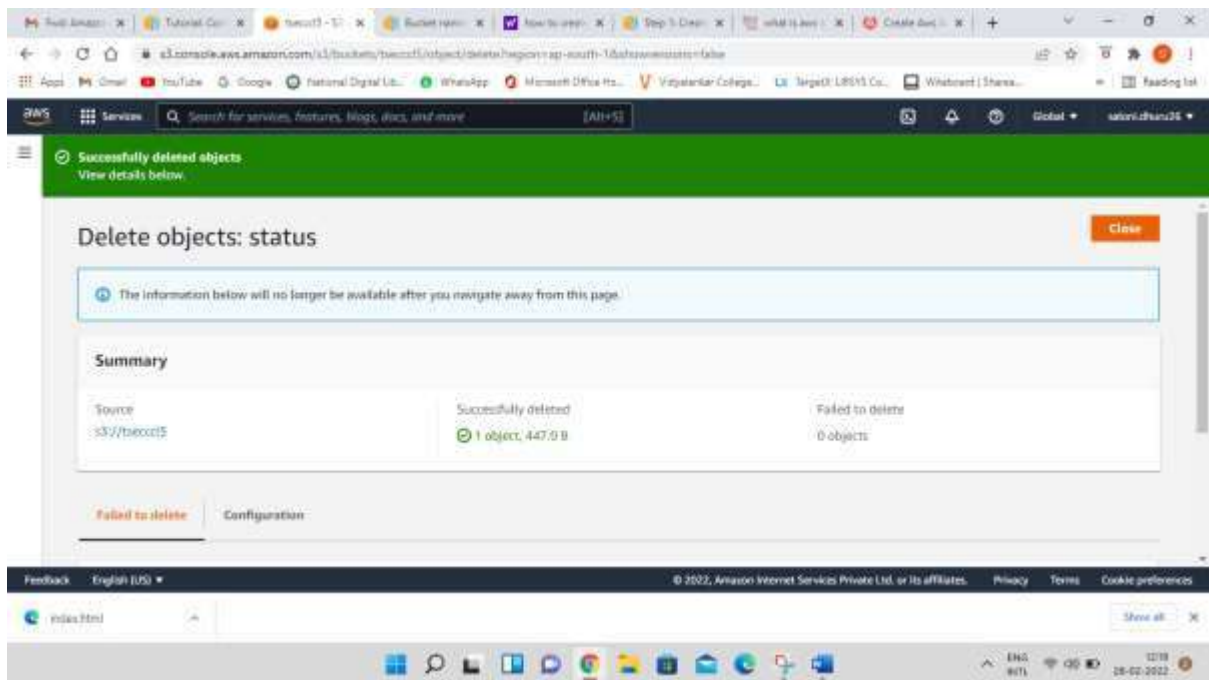
Step-17: Now for delete files click on checkbox of your file and then click on **Delete** Button



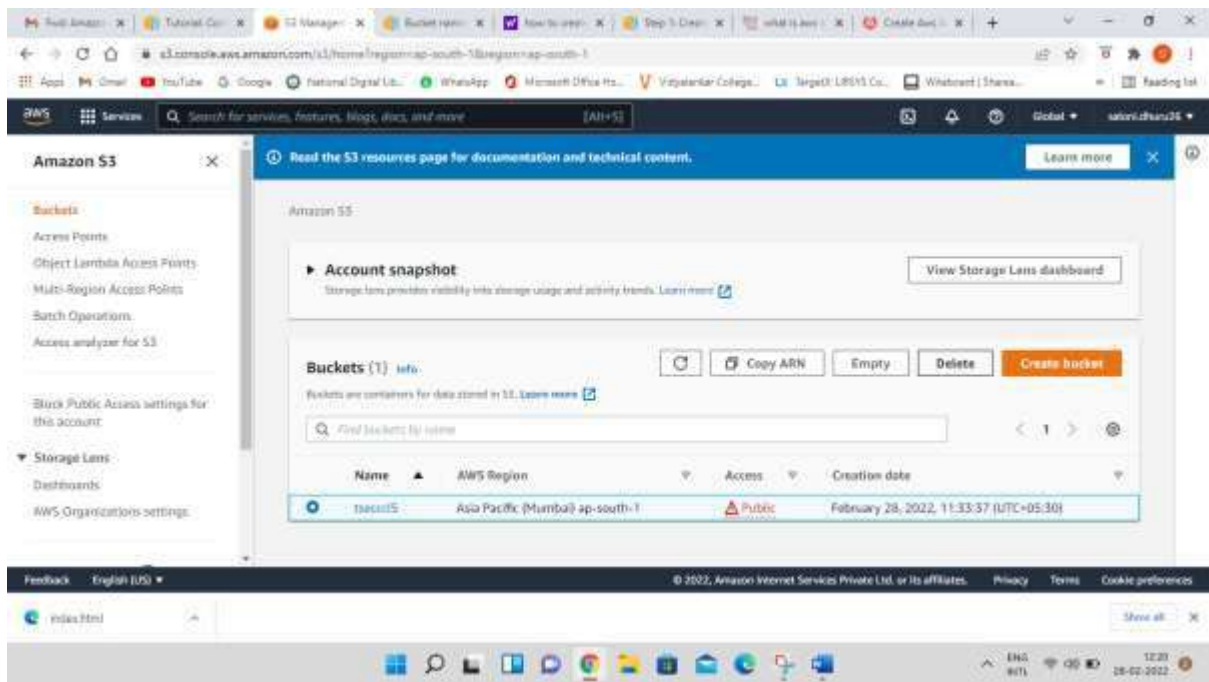
Write permanently delete and click on delete object button



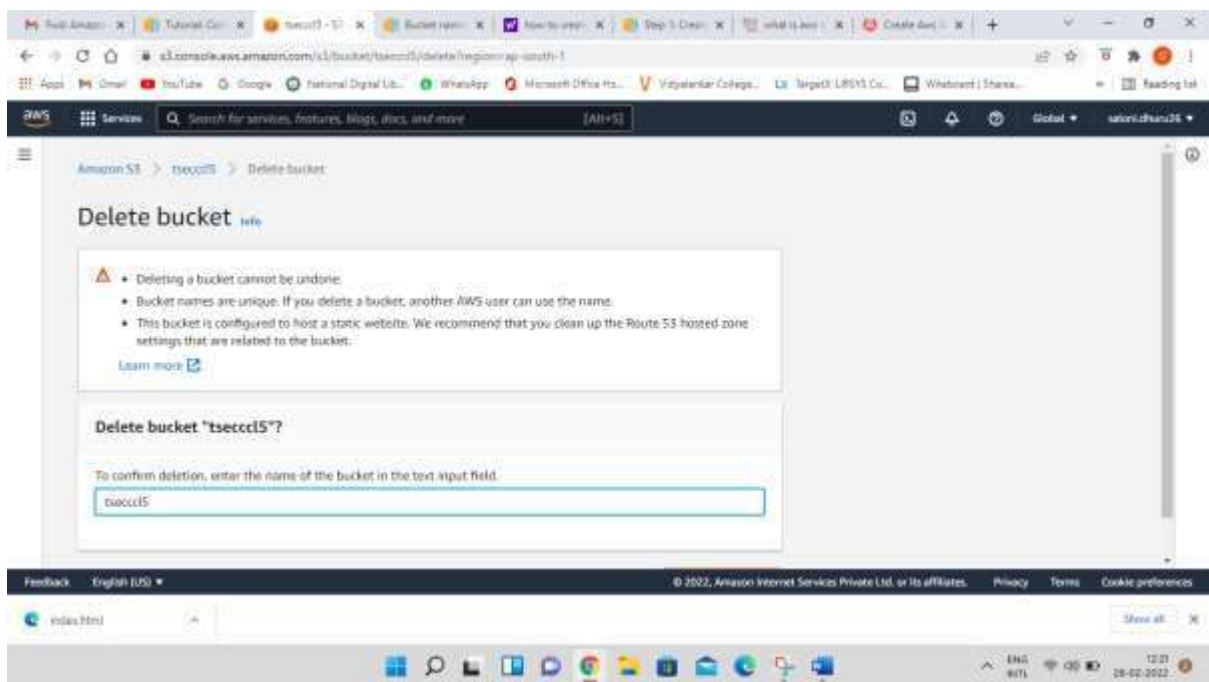
Now click on close button



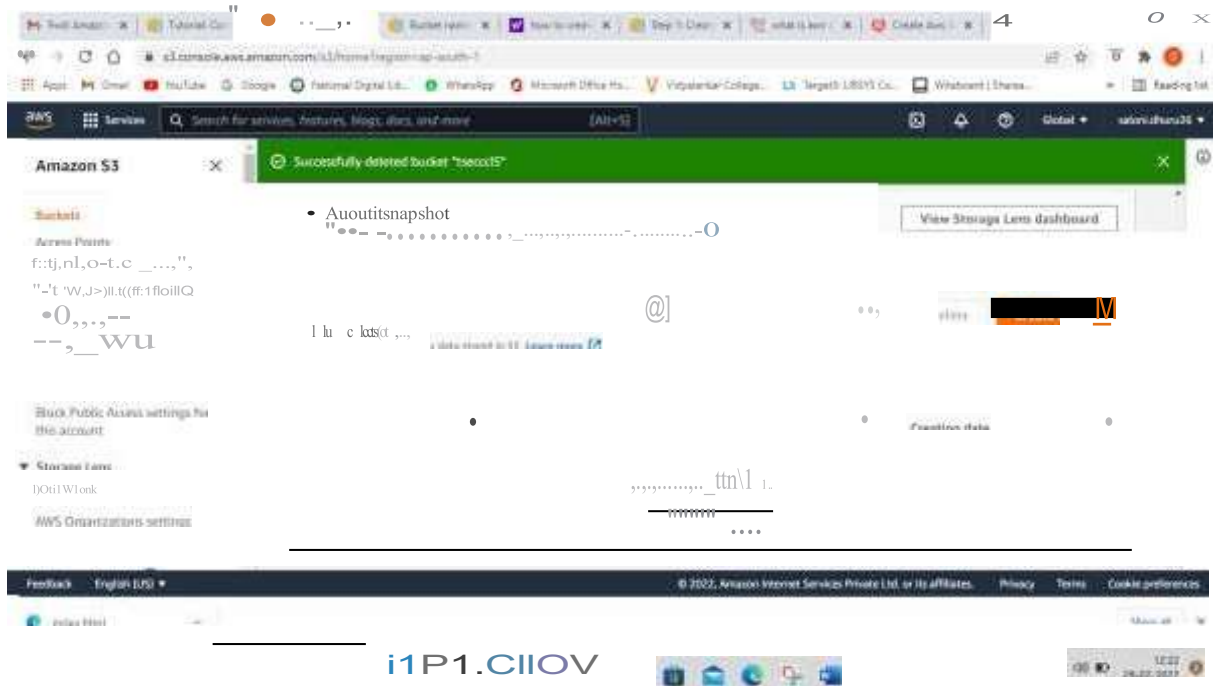
Step-18: now come to Amazon S3 tab and select your bucket and then click on delete button



Write down your bucket name in delete bucket tab and click on delete button at bottom right



You can see that the bucket is deleted



PART B
(PART B: TO BE COMPLETED BY STUDENTS)

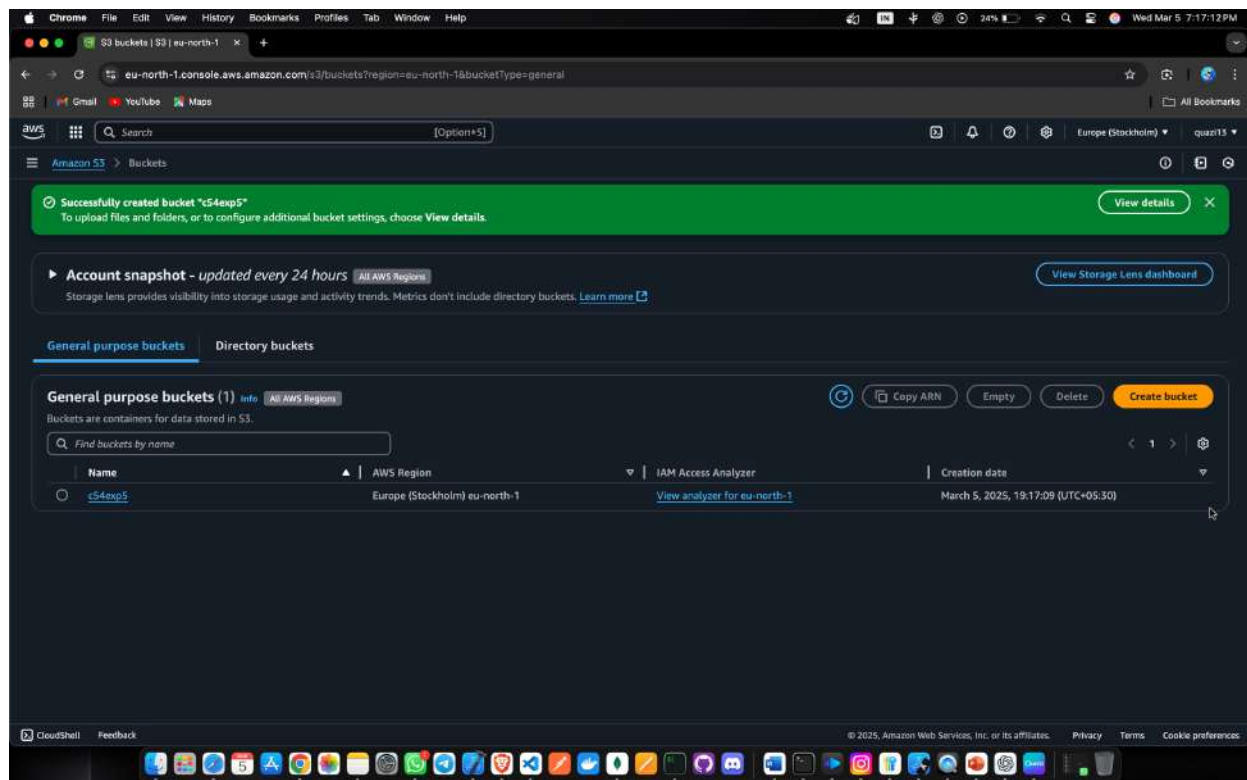
(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no ERP access available)

Roll No.C54	Name: Quazi Md Moinuddin
Class :TE-C	Batch :C3
Date of Experiment:	Date of Submission:5-2-25
Grade :	

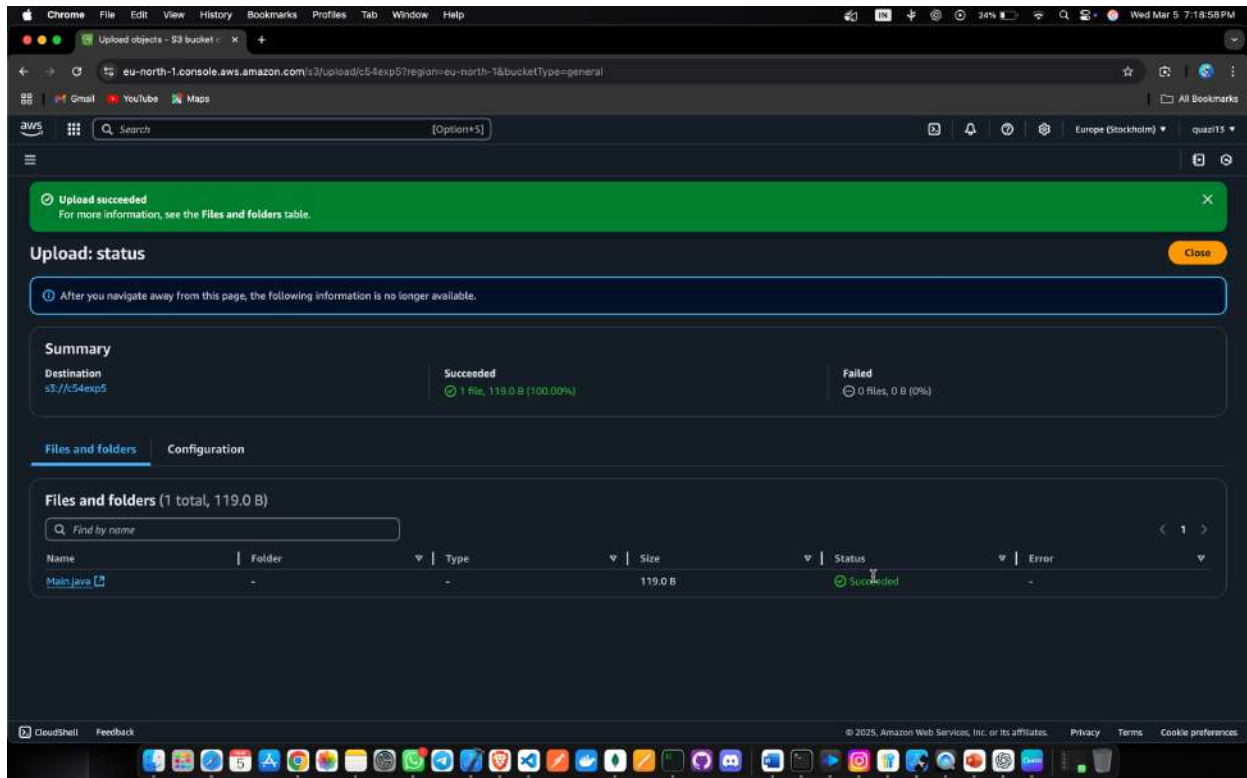
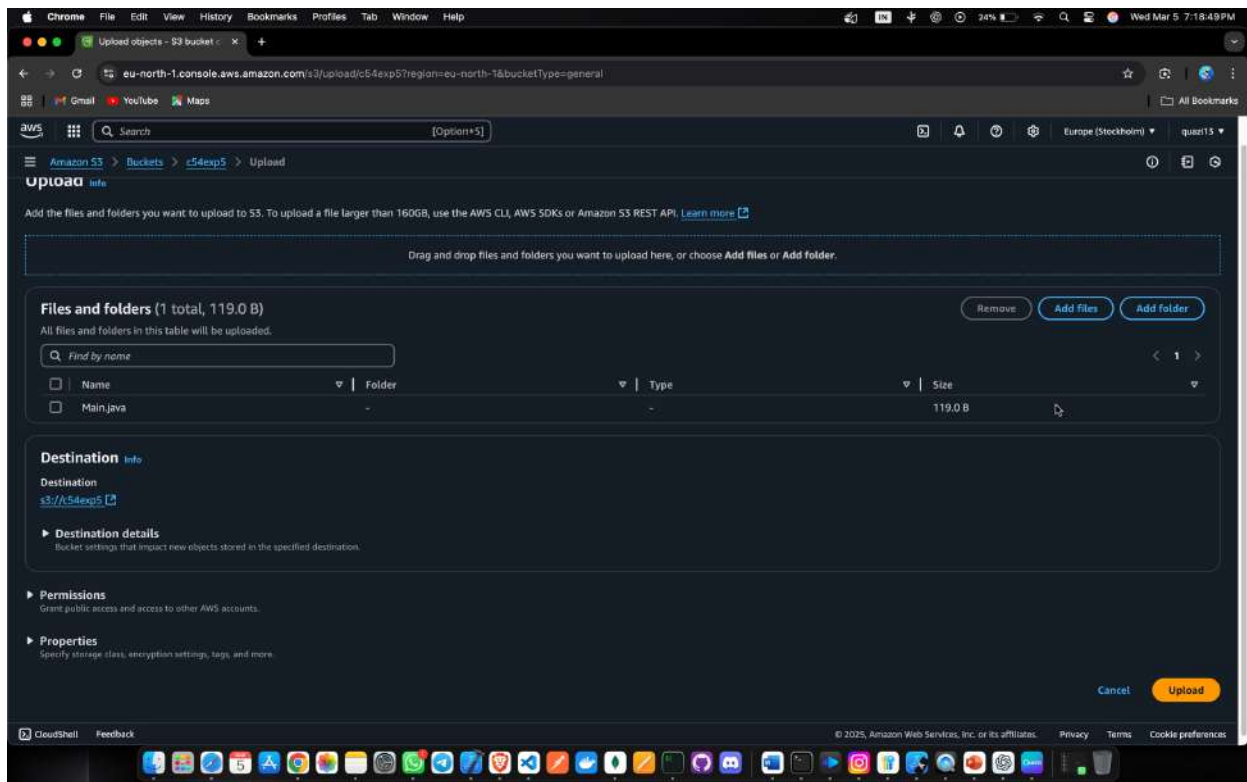
B.1 Question of Curiosity:

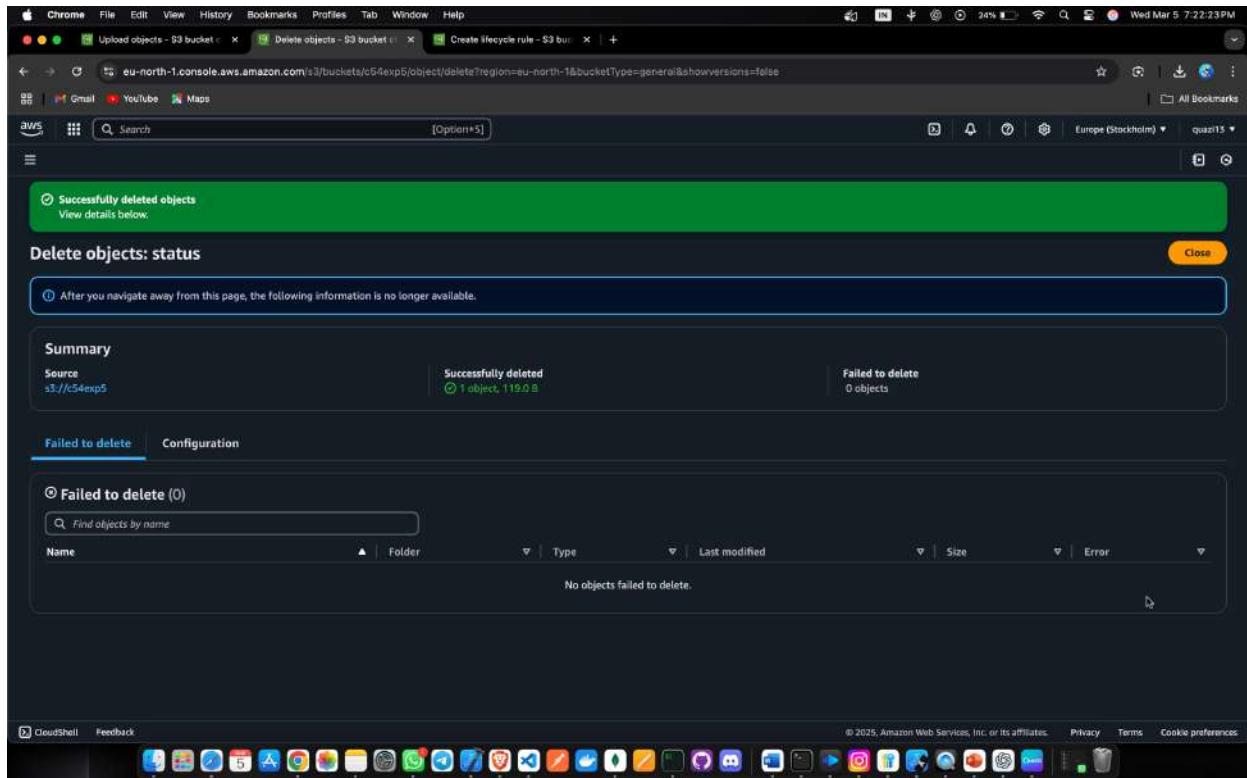
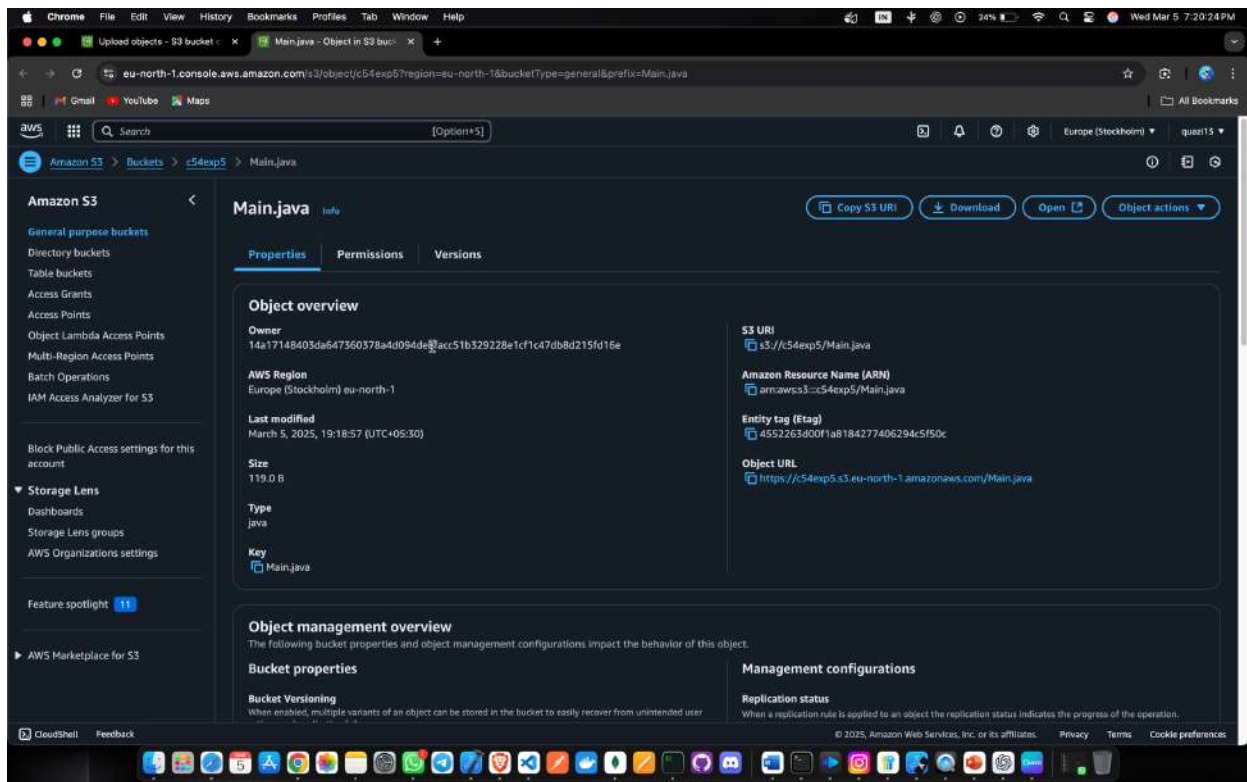
(To be answered by student based on the practical performed and learning/observations)

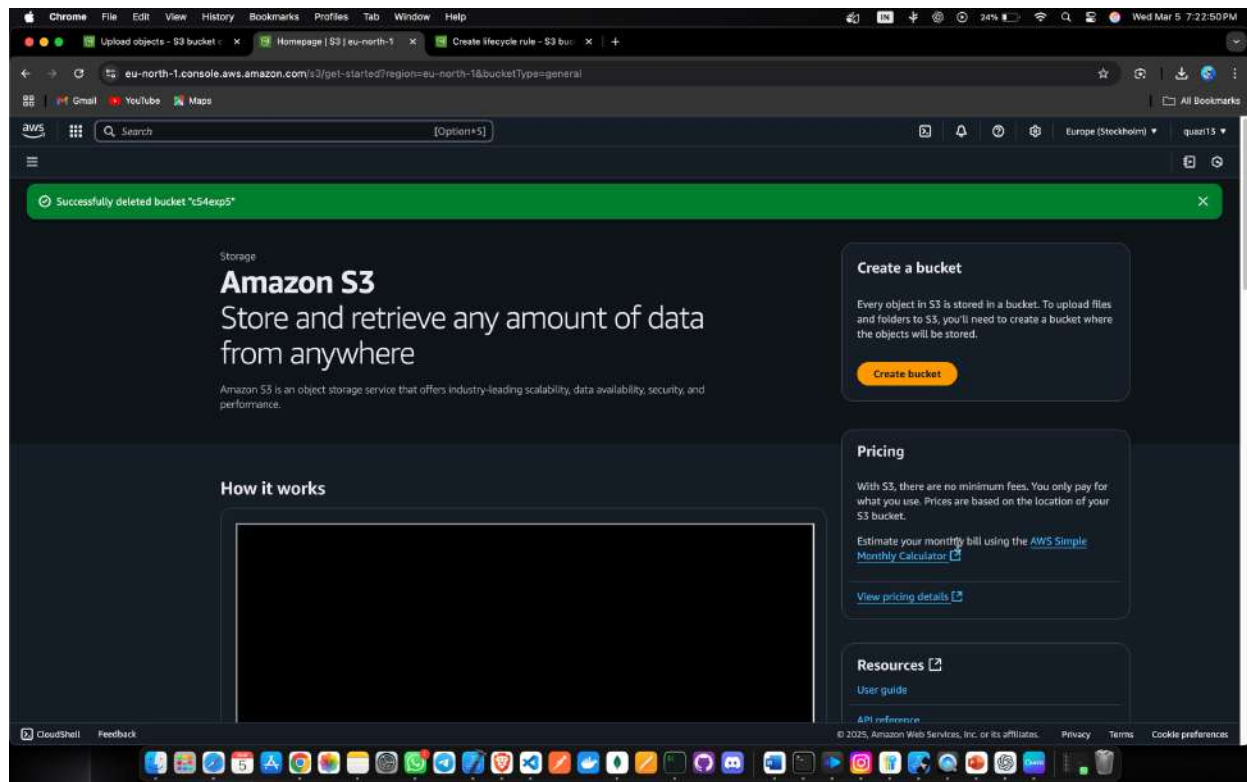
Q.1: Create Bucket using AWS S3 service (Add stepwise screenshots of the same)



Q2: Add Objects to Bucket created (Add stepwise screenshots of the same)







Q3: Compare Google drive with AWS S3

AWS S3	AWS EBS
The AWS S3 Full form is Amazon Simple Storage Service	The AWS EBS full form is Amazon Elastic Block Store
AWS S3 is an object storage service that helps the industry in scalability, data availability, security, etc.	It is easy to use.
AWS S3 is used to store and protect any amount of data for a range of use cases.	It has high-performance block storage at every scale.
AWS S3 can be used to store data lakes, websites, mobile applications, backup and restore big data analytics. , enterprise applications, IoT devices, archive etc.	It is scalable.
AWS S3 also provides management features.	It is also used to run relational or NoSQL databases.

B.2 Conclusion:

(Students must write the conclusion as per the attainment of individual outcome

listed above and learning/observation)

In conclusion, experiment provided a comprehensive understanding of storage services in AWS, particularly focusing on Amazon S3. Through practical implementation, students gained insights into the features and use cases of AWS storage solutions, facilitating their ability to deploy Infrastructure as a Service (IAAS) models in cloud computing. This experiment underscored the importance of storage management in cloud environments and equipped students with essential skills for leveraging AWS storage services effectively.